

Network Analysis

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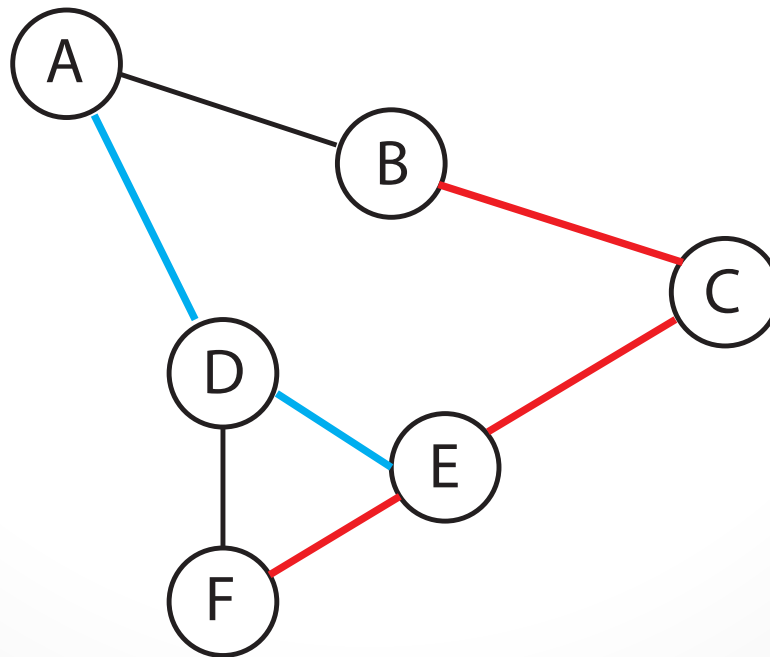
Descriptive Statistics

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Shortest Paths

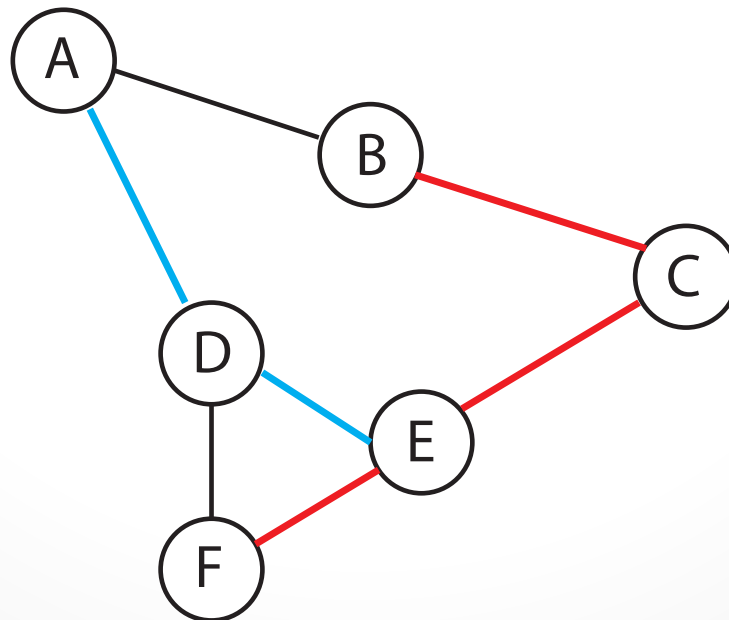
(Geodesic Distances)

- The **geodesic or graph distance** between two vertices is the length of the shortest path from one vertex to the other.
- For directed graph, must be a directed path



Diameter/Eccentricity

- **Graph Diameter:** Largest Geodesic Distance in the whole network
- **Eccentricity of a node:** Distance to furthest node from that node.



Density of Graph

- The density of a graph measures how interconnected the nodes are.
- Simply the proportion of possible edges that actually exist in the graph.
- Let E be the number of edges in the graph, N the number of vertices. The density, Δ , is then:

$$\text{➤ } \Delta = \frac{2E}{N(N-1)} \quad \text{For undirected graphs}$$

$$\text{➤ } \Delta = \frac{E}{N(N-1)} \quad \text{For directed graphs}$$

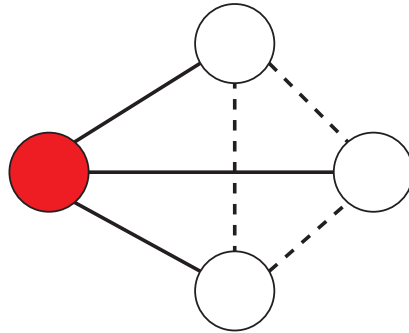
Density of Graph

- Several ways to consider density for a weighted graph depending on application.
- Some use average strength of edges.

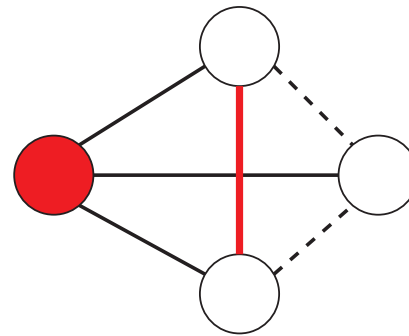
Clustering Coefficient

- The **clustering coefficient** of a node is a measure of the extent to which its neighbors are also neighbors of each other.
- Measures **Transitivity**: if A is connected to B and B is connected to C what is the probability that A is connected to C?
- Ratio of number edges existing between neighbors to those that could possibly exist.

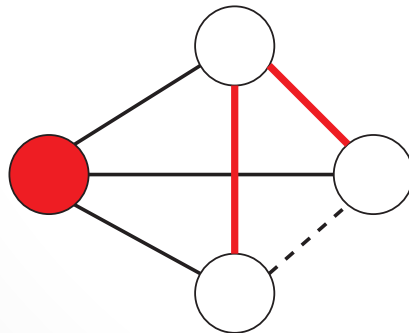
Clustering Coefficient



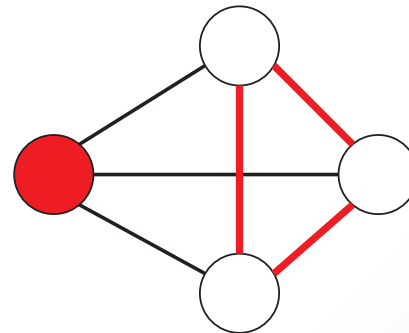
$$c=0$$



$$c=1/3$$



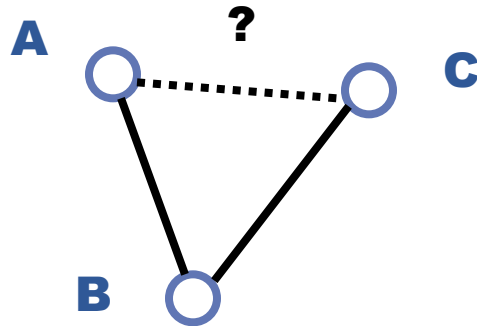
$$c=2/3$$



$$c=1$$

Clustering Coefficients for Entire Network

- Measure the transitivity of the entire network – does the transitive property hold most of the time?



Clustering Coefficients for Entire Network

- Network Average Clustering Coefficient: Simply average the clustering coefficient for each node.
- Global Clustering Coefficient:

$$C = \frac{3 \cdot \text{number of triangles in graph}}{\text{number of connected triples of vertices}}$$

- Connected triple is 3 vertices joined by 2 edges.
- Each triangle makes 3 connected triple.

Centrality

- In general, **centrality** is a measure of how important a vertex is in a graph.
- There are many ways to define importance.
- Common measures of centrality:
 - Degree Centrality
 - Betweenness Centrality
 - Closeness Centrality
 - Eigenvector Centrality
- Usually normalized in $[0,1]$ for comparison across networks

Degree Centrality

- Node with highest degree is most important
- Measures the exposure of vertex to others in network
- Degree centrality for vertex v is $C_D(v) = d_v / (n-1)$
 - d_v =degree of vertex v
 - n =number of vertices in network
- Max value is 1 (if vertex is connected to all other vertices).
- Local measure – can be deceiving.

Betweenness Centrality

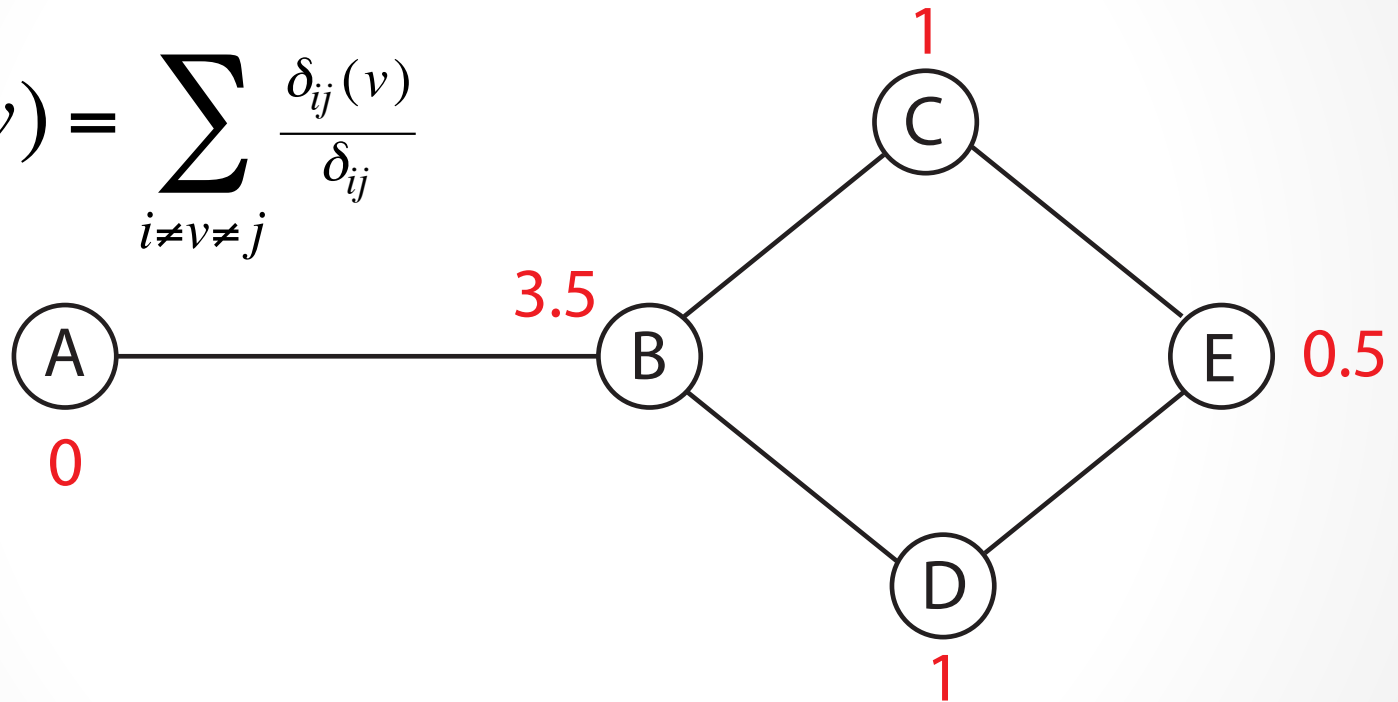
- Sum of proportions of shortest paths between 2 nodes that pass through the node of interest.
- Measures control of a node on communication between other nodes.

$$C_B(v) = \sum_{i \neq v \neq j} \frac{\delta_{ij}(v)}{\delta_{ij}}$$

- δ_{ij} =number of shortest paths between nodes i and j
- $\delta_{ij}(v)$ =number of shortest paths between i and j that go through node v
- Can also consider *edge betweenness centrality* using paths that include a given edge.

Betweenness Centrality

$$C_B(v) = \sum_{i \neq v \neq j} \frac{\delta_{ij}(v)}{\delta_{ij}}$$



Note: Be sure to use Brandes' Algorithm to compute betweenness! Developed in 2001, $O(mn)$ for unweighted graphs and $O(n^2 \log n + mn)$ for weighted graphs

Betweenness Centrality

$$C_B(v) = \sum_{i \neq v \neq j} \frac{\delta_{ij}(v)}{\delta_{ij}}$$

- Normalize each value by the maximum possible centrality score = the number of pairs of nodes excluding the given node.
 - For directed graph, normalize by $(n-1)(n-2)$
 - For undirected graph, normalize by $(n-1)(n-2)/2$
- Normalization allows for comparison across networks.

Closeness Centrality

- Measures how quickly information will spread from one node to all other nodes.
- Node is important if close to all other nodes.
- **Farness** defined as the sum of distances to all other nodes. **Closeness** is then the inverse of farness.

Closeness Centrality

$$C_c(v) = \frac{1}{\sum_{u \neq v} d(u, v)}$$

- Normalize by maximum possible score, which would be $1/(n-1)$ when the node is connected to all others.
 - i.e. multiply by $(n-1)$ to normalize.
- Only defined for connected graphs

Comparing 3 Centralities

- The centrality measures should be correlated.
- If not, that might mean something interesting.

	Low Degree	Low Closeness	Low Betweenness
High Degree		In cluster far from rest of network	Connections are redundant. Communication bypasses node
High Closeness	Key player, tied to important, active alternatives		Prob. multiple paths in network. Near many, just like everyone else
High Betweenness	A few edges are crucial for network flow	RARE. Node monopolizes ties from few to many	

Eigenvector Centrality

- The previous definitions of centrality have one short fall: They don't take into account the importance of your contacts.
- If you want to get someone hired, it's better to be connected to Dr. Rappa and Kathy Green than to Andrea Villanes or Matthew Austin.
- So previous measures don't necessarily measure ***influence***.

Eigenvector Centrality

- What if each node's centrality was the sum of the centralities of the nodes to which it is connected?
- Let $\mathbf{x}$ be the vector of centralities.
- Let $\mathbf{A}$ be a binary adjacency matrix.
- Then, $\mathbf{Ax} = \mathbf{x}$

Eigenvector Centrality

$$\mathbf{x} = \mathbf{A}\mathbf{x}$$

- Of course, this may not always be possible, so we'll add in a normalizing constant, λ :

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

- Look familiar??

Eigenvector Centrality

- But which eigenvalue??
- We want the centralities to be positive.
- The **first** eigenvector of an adjacency matrix is guaranteed to have all positive elements.
 - (Perron-Frobenius Theorem of Nonnegative Matrices)
- Google's PageRank is a variant of eigenvector centrality, using the hyperlink structure of the net.

Centralization of Network

- How much variation in centrality scores?
- How central is maximum centrality compared to others?
- Freeman's formula for **degree centralization**:

$$C_D = \frac{\sum_{i=1}^n [C_D(\text{max}) - C_D(i)]}{(n-1)(n-2)}$$

- Can be adopted for other types of centrality just by changing denominator and centrality function.

Some statistical issues with Network data

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Snowball sampling

- Data often collected via snowball sampling.
 - Select few individuals
 - Follow their ties/links to new individuals
 - Follow ties/links of new individuals
 - ...Not so random...
- Advantages
 - Great for hidden populations
 - Or when trust needed to collect info
- Disadvantages
 - Bias! towards initial set of individuals.
 - Potential for inaccurate referrals
 - Most imp: Doesn't tell you about true population distribution.

Independence

What independence??

Introduction to Gephi

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(and campaign finance)